Course Outcome

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

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End Term Examination, Odd Semester 2022-23
B.Tech (CSE), II Year, III Semester
B.Tech (CCV, DA, IIOT, AIML, CSF), III Year, V Semester
Software Engineering (BCSC0009)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt AllQuestions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Give the parametric comparison of Waterfall, Spiral and Prototyping model.	4	1	An	C
2	Suppose you are working on Requirement Analysis & Specification phase of a project. you are in need to create a joint team of customers and developers just to understand the expectations and propose a set of requirements. Now identify the best requirement elicitation technique in this case & what guidelines will you follow.	4	2	An	F
3	What is Module Coupling? Explain any four type of Coupling with the help of an example.	4	5	U	C
4	Consider a program for determining the Previous Date. Its input is a triple of day, month and year with the values in the range January $\leq$ month $\leq$ December $1 \leq$ day ≤ 30 $1950 \leq$ year ≤ 2075 Design the Robustness test cases for the same.	4	6	E	P
5	Consider a program given below for the selection of the largest of numbers	4	4	E	P

```
main()
   {
     float A,B,C;
     printf("Enter three values\n");
     scanf("%f%f%f",  &A,&B,&C);
     printf("\n Largest value is");
     if (A>B)
          {
               if(A>C)
                    printf("%f\n",A);
               else
                    printf("%f\n",C);
          }
          else
          {
               if(C>B)
                    printf("%f\n",C);
               else
                    printf("%f\n",B);
          }
    }
```

Draw the flow graph and DD graph. Calculate the Cyclomatic Complexity by all the methods.

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What is the difference between Reverse Engineering and Re-engineering? Explain with diagram.	3	7	U	F
7	A software project has development effort of 500 PM. It is assumed that 10% code will be modified per year. Some of the cost multipliers are given as: 1. Required software Reliability (RELY) : high= 1.15 2. Date base size (DATA) : high= 0.86 3. Analyst capability (ACAP) : high= 0.82 Other multipliers are nominal. Calculate the Annual Maintenance Effort (AME).	3	7	An	P

8	An authorized User can withdraw or deposit the money using an ATM machine. Create sequence diagram by showing all the necessary steps of ATM machine. Also discuss the purpose of creating sequence diagram.	3	4	An	F
9	Differentiate between • Alpha and Beta Testing • Functional Testing and Structural Testing • Validation and Verification	3	6	R	F
10	List the difference of CMM and ISO9001. Explain SEI-CMM Level.	3	7	R	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Why is maintenance programming becoming more challenging than new development? Explain the steps of software maintenance with help of a diagram. What is ripple effect? Enlist various categories of maintenance.	5	6	U	F
12	Consider the program for the selection of the largest of numbers in (Section-A Question-5). Compute the following- 1. Vocabulary of the program 2. program length 3. Program volume 4. Program level 5. potential volume	5	4	E	C
13	Consider a simple program to classify a triangle. Its inputs is a triple of positive integers (say x, y, z) and the date type for input parameters ensures that these will be integers greater than 0 and less than or equal to 100. The program output may be one of the following words: [Scalene; Isosceles; Equilateral; Not a triangle]. Identify the equivalence class test cases for output and input domain.	5	6	E	P

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End-Term Examination, Odd Semester 2022-23

B.Tech. CSE, II & III: Year, III & V: Semester

BCSE-0902: Smart Industrial Connectivity

Time: 1.5-hour **Maximum Marks: 50**

SECTION A

Note: Attempt All Questions. **30 x 1 =30 Marks**

1. What is the role of Cloud in smart grid architecture of IoT?
a) Security
b) Collect data
c) Manage data
d) Store data

2. What is the component of an IoT system that executes a program?
a) A sensor
b) A microcontroller
c) An actuator
d) A digital to analog converter

3. Which programming language is used by Arduino IDE IoT software for writing codes?
a) Python
b) Java
c) C/C++
d) JavaScript

4. What is the full form of IDE in Arduino IDE IoT software?
a) Intra Defence Environment
b) Intra Development Environment
c) Integrated Development Environment
d) Integrated Deployed Environment

5. Which of the following is not an actuator in IoT?
a) Stepper motor

b) A fan
c) An LED
d) Arduino

6. What is the use of PWM signals in IoT development boards?
a) They are used by sensors to have analog input
b) They are used by sensors to have digital input
c) They are used by actuators to have analog input
d) They are used by actuators to have digital input

7. Which of the following is true about Arduino IoT devices?
a) They are open-source software
b) They can only read analog inputs
c) They have their own operating systems
d) They don't have pre-programmed firmware

8. Which of the following is not a feature of the Raspberry PI model B IoT device?
a) It has 256 MB SDRAM
b) It has a single USB connector
c) It has its own operating system
d) It has an Ethernet port

9. Which of the following processors is used in the Raspberry PI 3 IoT device?
a) Broadcom BCM2711
b) Broadcom BCM2837
c) Broadcom BCM2838
d) Intel 8085

10. What IoT collects?
a) Device data
b) Machine generated data
c) Sensor data
d) Human generated data

11. What is the role of Cloud in smart grid architecture of IoT?

a) Security
b) Collect data
c) Manage data
d) Store data

12. What is the component of an IoT system that executes a program?
a) A sensor
b) A microcontroller
c) An actuator
d) A digital to analog converter

13. What is the full form of IDE in Arduino IDE IoT software?
a) Intra Defence Environment
b) Intra Development Environment
c) Integrated Development Environment
d) Integrated Deployed Environment

14. Which of the following is used to capture data from the physical world in IoT devices?
a) Sensors
b) Actuators
c) Microprocessors
d) Microcontrollers

15. Which of the following is false about the IoT components?
a) A light sensor (photoresistor) is an analog sensor
b) A microphone is a digital sensor
c) A push button is a digital sensor
d) A keyboard is a digital sensor

16. What is the full form of IIoT?
a) Index Internet of Things
b) Incorporate Internet of Things
c) Industrial Internet of Things
d) Intense Internet of Things

17. Which of the following is not a fundamental component of an IoT system?
a) Sensors
b) Connectivity and data processing
c) User interface
d) Transformer

18. SPI is described as an Asynchronous serial interface.
a) True
b) False

19. Which of the following is not an IoT platform?
a) Amazon Web Services
b) Microsoft Azure
c) Salesforce
d) Flipkart

20. Which of the following is not a basic element within the microprocessor?
A. Microcontroller
B. Arithmetic logic unit (ALU)
C. Register array
D. Control unit

21. HTTP expands?
a) Hypertext Transfer Protocol
b) HyperTerminal Transfer Protocol
c) Hypertext Terminal Protocol
d) HyperTerminal Text Protocol

22. What is IoT?
a) network of physical objects embedded with sensors
b) network of virtual objects
c) network of objects in the ring structure
d) network of sensors

23. What is ESP8266?

a) WIFI module
b) Sensor
c) Board
d) USB cable

24. Does SPI have/has a single master?
a) True
b) False

25. Which proximity sensor detects positioning of an object?
a) Inductive Proximity Sensor
b) Capacitive Proximity Sensor
c) Optical Proximity Sensor
d) Magnetic Proximity Sensor

26. Rx and Tx stands for:
A. Register and Transistor
B. Reload and Transmitter
C. Read and Transfer
D. Receiver and Transmitter

27. I2C will address large number of slave devices.
a) True
b) False

28. Which of the following needs a clock?
a) Only Asynchronous
b) Only synchronous
c) Both synchronous and Asynchronous
d) Sometimes Synchronous

29. Resolution is expressed in _____
a) Bytes
b) Bits
c) Word
d) Nibble

30. Master transmits means _____
a) Master node is sending data to a slave
b) Master node is receiving data from slave
c) Slave node is transmitting data to master
d) Slave node is sending data to master

SECTION B

Note: Attempt All Questions. **40 x 0.5 =20 Marks**

31. Which sensor is LM35?
a) Pressure sensor
b) Humidity sensor
c) Temperature sensor
d) Touch sensor

32. DHT11 is _____ sensor.
a) Proximity sensor
b) Humidity sensor
c) Touch sensor
d) Pressure sensor

33. Size of EEPROM available in Arduino UNO:
A. 1 KB
B. 2 KB
C. 512 Byte
D. 128 Byte

34. Two wire interface is also called as _____
a) UART
b) SPI
c) I2C
d) USART

35. Which sensor can detect nearby objects?
a) Proximity sensor
b) Humidity sensor

c) Touch sensor
d) Pressure sensor

36. Relative humidity is a function of _____
a) Ambient temperature
b) Water vapor pressure
c) Ambient temperature and Water vapor pressure
d) Dryness

37. GPS modules like SIM900/800 use which protocol?
a) UART protocol
b) USART protocol
c) SPI protocol
d) I2C protocol

38) Which sensors measure the moisture level using humidity?
a) Capacitive Sensor
b) Resistive Sensor
c) Thermal Conductivity
d) Both resistive and conductive

39. Which is false about IoT devices?
A. IoT devices use the internet for collecting and sharing data
B. IoT devices need microcontrollers
C. IoT devices use wireless technology
D. IoT devices are completely safe

40. How much volts do the HC-SR04 ultrasonic distance sensor work with?
A. 1.2V
B. 3.3V
C. 9V
D. 5V

41. Materials with lots of free electrons are called:
A. Conductors
B. Insulators

C. Semiconductors
D. Filter

42. What is the main purpose of WoT (Web of Things) in the IoT?
a. Improve the usability and interoperability
b. Reduce the security
c. Complex the development
d. Increase the cost

43. An IoT network is a collection of _____ devices.
a. Signal
b. Machine to Machine
c. Interconnected
d. Network to Network

44. What is the full form of the LPWAN?
a. Low Protocol Wide Area Network
b. Low Power Wide Area Network
c. Long Protocol Wide Area Network
d. Long Power Wide Area Network

45. What is the main purpose of WoT (Web of Things) in the IoT?
a. Improve the usability and interoperability
b. Reduce the security
c. Complex the development
d. Increase the cost

46. Which communication provides the fastest data transmission?
A. RS-232
B. Parallel
C. Fiber-optic cable
D. None of the above

47. In which frequency range does the RFID system operate?
a.) High- Frequency
b.) Low-Frequency
c.) Ultra-frequency

d.) All of the above

48. What is the range of RFID tags which are using the high frequency?
a.) Up to 10 cm
b.) Up to 1 meter
c.) Up to 60 cm
d.) Up to 20 cm

49. For high frequency and low frequency RFID tags the working principle is based on
a.) Inductive Coupling
b.) Electromagnetic Coupling
c.) both a & b
d.) none

50. What is the standard form of UART?
a.) Universal asynchronous receiver-transmitter
b.) United asynchronous receive-transmit
c.) Universal All receiver-transmit
d.) None

51. What are the strengths of RFID?
a.) Advanced Technology
b.) Small Size, easy to use
c.) Has high Memory capacity
d.) None

52. Which one of the following is reusable?

a.) RFID
b.) Barcode
c.) both a & b
d.) None

53. Which one of the following operates at higher frequencies?
a.) Active
b.) Passive

c.) both a & b
d.) None

54. What is a different kind of radio frequency identification tag?
a.) Active
b.) Semi-Passive
c.) Passive
d.) All of the above

55. What is Eclipse?
a) IDE
b) Software
c) GNU
d) Code base

56. How many USB ports are present in Raspberry Pi 3?
a) 5
b) 2
c) 4
d) 3

57. What is the full form of SPI?
a) Series Parallel Interface
b) Serial Peripheral Interfacing
c) Series Program Intercommunication
d) Serial Proxy Immune

58. How is the power supply done to RPi?
a) USB connection
b) Internal battery
c) Charger
d) Adapter

59. Wi-Fi is not present in which of the following models?
a) Raspberry Pi3
b) Raspberry Pi Zero WH
c) Raspberry Pi Zero W

d) Raspberry Pi Zero

60. What do we use to connect TV to RPi?
a) Male HDMI
b) Female HDMI
c) Male HDMI and Adapter
d) Female HDMI and Adapter

61. Does RPi have an internal memory?
a) True
b) False

62. What does an Eclipse IDE allow?
a) Editing the program
b) Compiling the program
c) Both editing and compiling
d) Editing, building, Debugging, Compiler

63. Capacitive RH sensors dominate _____ measurements.
a) Atmospheric
b) Process
c) Both atmospheric and process
d) Neither atmospheric nor process

64. What is a humidity sensor?
a) Hygrometer
b) Gyroscope
c) Seismoscopes
d) Sundial

65. Units for Humidity is _____
a) grams/m3
b) % by volume
c) PPMV
d) %

66. Humidity sensors are divided into how many types and depending on what?
a) 2 types depending on output
b) 2 types depending on measurement units
c) 3 types depending on measurement units
d) 3 types depending on output

67. Electronic type hygrometers are divided into how many types?
a) 5 types
b) 3 types
c) 4 types
d) 2 types

68. How many components does the RFID system consist of?
a.) One
b.) Two
c.) Three
d.) Four

69. For ultra-high frequency RFID tags the working principle is based on _____
a.) Inductive Coupling
b.) Electromagnetic Coupling
c.) both a & b
d.) none

70. Which proximity sensor detects metal objects?
a) Capacitive Proximity Sensor
b) Magnetic Proximity Sensor
c) Ultrasonic Proximity Sensor
d) Inductive Proximity Sensor

Course Name: **OBJECT-ORIENTED PROGRAMMING SYSTEM**

Course Outcome

CO1- Understand the concepts of OOPS and Java.
CO2- Implement various types of arrays.
CO3- Implement Exception handling, wrapper classes, and collections
CO4- Understand abstract class and interface
CO5- Implement JDBC and its classes
CO6- Implement multithreading and regular expression

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End Term Examination, Odd Semester 2022-23

B. Tech. (CSE/CCV/DA/IIOT/AIML/CSF), II- Year, III- Semester

BCSC-1002 OBJECT-ORIENTED PROGRAMMING SYSTEM

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Read the question carefully.
2. Don't Overwrite
3. Write the complete code in one place neatly
4. Maintain appropriate indentation while writing code in java if needed.
5. Commenting code is optional.
6. **Write the output of each program.**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

Sr. No.	Detail of Question	Marks	CO	BL	KL
1	Explain features of Object Oriented Programming. Describe at least 6 principles of Object Oriented Programming in detail.	4	CO1	R	C
2	Define array. Explain types of arrays. Write a program to print the sum and average of the array.	4	CO2	U	P
3	Differentiate between abstract class and interface. Also, illustrate the suitable code for each.	4	CO4	An	D
4	Create a collection to store objects of the Book class by using ArrayList. The Book objects contain bookname. a) Insert 5 booknames in the ArrayList. b) Display all the contents of the book class using the iterator methods and object. c) Delete the last bookname from the ArrayList. Write the output of the program.	4	CO3	A	M
5	Create an abstract class 'Bank' with an abstract method 'getBalance' $100, $150 and $200 are deposited in banks A, B and C respectively. 'BankA', 'BankB' and 'BankC' are subclasses of class 'Bank', each having a method named 'getBalance'. Call this method by creating an object of each of the three classes. Or Explain the lifecycle of multithreading with the help of a diagram. Write the program to create a thread to print the factorial of a number using the runnable interface.	4	CO4	A	PC

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Define the following terms. Also, write the statements to implement them. a) Driver Manager b) Connection c) Statement d) ResultSet	3	CO5	U	S
7	Create the class to find the index position of the pattern "o" in a string "Hello How are you" using pattern and matcher classes methods.	3	CO6	A	PC
8	What is the use of wrapper classes in Java? Write the statement of code for the following: a) Primitive data type to Object type b) Object to Primitive data type.	3	CO3	U	S
9	Write the short note on: a) This b) Sleep() c) Thread Priorities	3	CO6	U	F

10	Predict the output of the following codes. Also, give an explanation.
	a)
	`abstract class A`
	`{ public void disp()`
	`{`
	`System.out.println("Concrete method of parent class");`
	`}`
	`abstract public void disp2();`
	`}`
	`class B extends A`
	`{`
	`public void disp2()`
	`{`
	`System.out.println("Overriding abstract method");`
	`}`
	`public static void main (String args[])`
	`{`
	`B obj = new B();`
	`obj.disp();`
	`}`
	`}`
	
	b)
	`public class TestOne`
	`{`
	`public static void main(String[] args) throws Exception`
	`{`
	`Thread.sleep(3000);`
	`System.out.println("sleep");`
	`}`
	`}`
		3	CO1	E	M

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	What is exception ? Why the exception occurred in program? What is the use of try, catch and finally statement give example? Create a class and define one method which throws ArithmeticException.	5	CO3	A	P
12	Write a program to setup JDBC and execute an SQL statement on database table "EMPLOYEE", with the fields Empid, EmpName, EmpSalary. Execute the following queries on database table. a) Insert 3 records in database table. Write the output of the java command prompt and MYSQL. b) Retrieve all the records from the database table. Write the output of the java command prompt and MYSQL.	5	CO5	A	P
13	Create a class named 'Member' having the following members: Data members (Name, Age, Phonenumber, Address and Salary.It also has a method named 'printSalary' which prints the salary of the members. Two classes 'Employee' and 'Manager' inherits the 'Member' class. The 'Employee' and 'Manager' classes have data members 'Designation' and 'Department' respectively. Now, assign name, age, phone number, address and salary to an employee and a manager by making an object of both classes and print the same.	5	CO1	C	PC

Course Name: BCSC1005, Computer Organization

Course Outcome

CO1: Understand the basics of digital computer system.
CO2: Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3: Understand the concepts of Combinational and Sequential circuits and their applications.
CO4: Understand the CPU architecture and organization.
CO5: Explain the basic concepts of pipelining.
CO6: Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7: Explain the function of memory hierarchy.
CO8: Determine the interface of CPU with input/output devices and their modes of transfer.

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End Term Examination, Odd Semester 2022-23
B. Tech. CS (ALL), II Year, III Semester
BCSC1005, Computer Organization

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Attempt all questions.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Perform in steps the multiplication of the numbers (-7) and (+4) by using Booth's algorithm	4	2	A	P
2	What are the essential component of a digital computer? Discuss briefly the function of each component by giving appropriate block diagram.	4	1	U	F
3	Explain PUSH and POP operations. Convert the following arithmetic expression into reverse polish notation and show the show the stack operations for evaluating it? 5+8*7+(4+3)+8*(5+3)	4	4	A	P
4	Differentiate hardwired and microprogrammed control unit by using appropriate figure? Also discuss its advantage and disadvantage.	4	6	U	C
5	What do you mean by cache memory? Discuss its different types. Differentiate write through and	4	7	U	F

	write back with respect to cache memory. Or What is instruction cycle? Draw the flowchart for instruction cycle.				

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Show the execution of complete instruction for Add R1,R3 where contents of Register R1 and R3 are added and result is stored in R3.	3	6	A	P
7	Draw Memory Pyramid. Explain the difference between various memories on the basic of Storage, Cost per bit, Access mode and Speed.	3	7	U	C
8	What do you mean by high-impedance state of buses? A computer uses RAM chips of 1024x1 capacity. How many chips are needed to provide a memory capacity of 1024 bytes? How many address lines are required to access 2048 bytes of memory?	3	7	A	P
9	What do you mean by Interrupt? Discuss different types of interrupt in detail.	3	8	U	C
10	What do you mean by pipeline processing? In a certain scientific computation, it is necessary to perform the arithmetic operation $A_i * B_i * (C_i * D_i)$ with a stream of numbers. Specify a pipeline configuration to carry out this task.	3	5	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	What is DMA controller? Explain in detail the Direct Memory Access (DMA) transfer in a computer system by giving appropriate figure.	5	7	U	C

12	A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. How many RAM and ROM chips are needed? Draw a memory-address map for the system. Give the address range in hexadecimal for RAM, ROM and interface.	5	7	A	P
13	What do you mean by virtual memory? How it is useful for executing any program. A virtual memory system has an address space of 8K words, a memory space of 4K words, and page and block sizes of 1K words. The following page reference changes occur during a given time interval. 5 1 3 2 3 7 5 2 1 2 3 1 4 8 6 Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is FIFO and LRU.	5	7	A	P

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment
CO2- Understand the basic of process management
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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End Term Examination, Odd Semester 2022-23
B. Tech. (CSE / EC with minor in CSE), II Year, III Semester
Operating Systems (BCSC0004)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No	Detail of Question	Marks	CO	BL	KL
1	State the Reader Write problem and discuss its solution using Semaphore.	4	4	U	C
2	What is convoy effect? Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds. Process Name / Arrival Time / Execution Time A / 0 / 6 B / 3 / 2 C / 5 / 4 D / 7 / 6 E / 10 / 3 Using the *shortest remaining time first* scheduling algorithm, calculate the average process turnaround time (in msec).	4	3	A	P
3	What is the main purpose of an operating system? Also, explain the advantages and applications of Multithreaded System?	4	1	U	C
4	Given 3 processes A, B and C, three resources x, y and z and following events, i) A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y Assume that requested resources should always be allocated to the request process if it is available. Draw the	4	5	An	P

	resource allocation graph for the sequences. And also mention whether it is a deadlock? If it is, how to recover the deadlock.				
5	What is seek time? Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive currently services a request at cylinder 143, and the previous request was at cylinder 125. the queue of pending request in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Starting from the current position, what is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests, for each of the following algorithms i) SSTF ii) C-SCAN.	4	7	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain Belady's Anomaly with an example. A system uses FIFO policy for page replacement. It has 4-page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?	3	7	An	P
7	What is paging? Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is	3	6	U	C
8	In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 16 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits	3	6	An	M

	that can be used for storing protection and other information in each page table entry?				
9	Differentiate between internal fragmentation and external fragmentation with example.	3	5	U	R
10	Explain the different methods for free space management with an example.	3	7	An	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
11	Discuss the segmentation memory management technique with an example? Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using- 1. First Fit Algorithm 2. Best Fit Algorithm 3. Worst Fit Algorithm	5	6	A	P
12	Explain different file allocation methods with their advantages and disadvantages.	5	7	R	C
13	Differentiate between virtual address and physical address. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 On a demand paged virtual memory system running on a computer system that main memory size of 3 pages frames which are initially empty. Find the number of page faults under the corresponding page replacements policy: LRU, FIFO and OPTIMAL.	5	6	An	M

Course Name: Computer Networks

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

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End Term Examination, Odd Semester 2022-23

B.Tech. (CSE/AIML/CCV/DA/CSF/IOT), 2nd Year, III Semester

BCSC0008 Computer Networks

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: ----------

Section – A

4 X 5 = 20 Marks

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL
1	How Go Back N –ARQ is differ from Selective repeat request. Explain with suitable example.	4	3	R	F
2	Write a note on guided transmission media? Draw neat diagram wherever required.	4	2	R	F
3	Distinguish 1 persistence, Non-persistence and P persistence CSMA	4	3	R	F
4	What is class full and class less IP addressing scheme. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?	4	5	A	P
5	How TCP is differing from UDP? Discuss connection Establishment process of TCP.	4	6	U	C

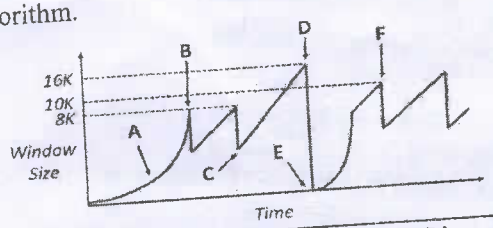
Section – B

3 X 5 = 15 Marks

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL
6	Explain Public Key and Private Key cryptography. In a RSA cryptosystem a particular A uses two prime numbers p = 13 and q =17 to generate her public and private keys. If the public key of A is 35.	3	7	A	P

then generate private key of A.

7	Name the event at A,B,C,D,E,F which occurs that causes the sender to decrease its window of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm.  	3	6	U	C
8	What is two node instability problem? Discuss the difference between distance vector and Link State Routing.	3	5	R	F
9	What is the purpose of a subnet mask? Is the subnet mask 255.255.0.255 valid for a Class A address? If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?	3	5	U	C
10	An IPv4 packet has arrived with the first 8 bits as shown:01000010. The receiver discards the packet. Why? Consider sending a 3,000-byte datagram into a link which has an MTU of 500 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are their characteristics	3	8	An	P

Section – C

5 X 3 = 15 Marks

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL
11	Consider an instance of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission. OR	5	6	A	C

	The following is a dump of a TCP header in hexadecimal format. 05320017 00000001 00000000 500207FF 00000000 a. What is the source port number? b. What is the destination port number? c. What is the sequence number? d. What is the acknowledgment number? e. What is the length of header? f. What is the type of segment? g. What is the window size?				
12	An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. What will be valid allocation of addresses to A and B? Explain your answer with proper justification.	5	5	An	C
13	Consider the following routing table at an IP router for each IP address in Group-I identify the correct choice of the next hop using the entries from the routing table below. Group - I A. 128.96.171.92 B. 128.96.167.87 C. 128.96.163.121 D. 128.96.165.121	5	5	A	P

Network No.	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.254.0	R3
0.0.0.0	Default	R4

Course Name: DISCRETE MATHEMATICS

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5. Analyze the validity of an argument using logical notation.
CO6. Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7. Understand the basic principles of sets, set equalities and operations in sets.
CO8. Apply counting principles to determine arrangements and solve recurrence relations.

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End Term Examination, Odd Semester 2022-23
B.Tech. CSE/CCV/DA/CSF/AIML/IIoT, II Year, III Semester
B.Tech. CCV/DA/CSF/AIML/IIoT, III Year, V Semester
BCSE 1010: DISCRETE MATHEMATICS

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Attempt all the questions of a section at one place.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Solve the following recurrence relation using Generating Functions: $y_r - 7y_{r-1} + 10y_{r-2} = 0, r \geq 2; y_0 = 2, y_1 = 1$	4	1	U	P
2	a) There are 5 gentlemen and 4 ladies to dine at a round table. In how many ways can they sit themselves so that no two ladies are together? b) What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five grades A, B, C, D, E?	2+2	2	U	F
3	Find the complement of each element of D_{42} where D_m denote the positive divisors of m ordered by divisibility. Also determine whether it is a lattice.	4	3	R	C
4	Find order of each element in the group $<\{1, -1, i, -i\}, * >$. Determine whether it is a cyclic group. If so, find all its generators.	4	4	U	C
5	Draw the graphs K_3, C_7, W_5 and $K_{2,3}$.	4	4	A	P

Section – B

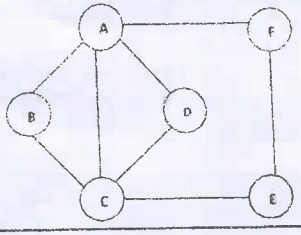
Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Write the converse, inverse and contrapositive of the statement: **"If it rains then I will bring umbrella."**	3	5	An	P

7	Obtain PCNF for the formula: $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$	3	5	A	F
8	Show that the set $Q - \{1\}$ of rational numbers other than 1 is an abelian group under the composition * defined as $a * b = a + b - ab$.	3	4	An	M
9	Rewrite the following argument using quantifiers, variables, and predicate symbols: a) All birds can fly. b) Some birds are mammals. c) Not all birds are quarrelsome.	3	6	R	C
10	Explain Planar graph with suitable example. Let G be a connected planar simple graph with 25 vertices and 60 edges. Find the number of regions in G.	1+2	4	A	M

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	**a)** Show that $((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow ((p \vee q) \rightarrow r)$ is a tautology. **b)** Show that **'s'** is a valid conclusion from the given premises: $p \rightarrow \neg q, q \vee r, \neg s \rightarrow p, \neg r$	2+3	5	An	M
12	Show that $<Z_5, +_5, x_5>$ is a field.	5	4	A	P
13	Define bipartite graph. Find whether the following graph is bipartite? Also find Euler Path/Circuit and Hamiltonian Path/Circuit in the following graph, if it contains. 	1+1+3	6	R	C

Course Name: **Engineering Mathematics-III**

Course Outcome:

CO 1: Apply method of least square for fitting of straight line

CO 2: Apply Chi-square test

CO 3: Use Laplace transform in solving differential equations

CO 4: Know the concept of analytic functions and its applications in engineering

CO 5: Understand bilinear transformation and its applications

CO 6: Use Cauchy's residue theorem in solving real integrals

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End Term Examination, Odd Semester 2022-23
Program- B. Tech. (CSE, CE, EE, ECE, EN) - II Year, III Semester
Subject Name: Engineering Mathematics-III
Subject Code: BMAS 1103

Time: 3 Hours **Maximum Marks: 50**

Note – Attempt ALL the sections.

Section – A

Attempt All Questions. $4 \times 5 = 20$ Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Fit a straight line to the following data: x: 0 1 2 3 4 y: 1 1.8 3.3 4.5 6.3 by the method of least squares.	4	1	A	P
2	Solve the differential equation by using Laplace transform: $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}\sin t,$ with $x(0) = 0,\ x'(0) = 1.$	4	3	U	C
3	A die is thrown 270 times and the results of these throws are given below: No. appeared on the die: 1 2 3 4 5 6 Frequency: 40 32 29 59 57 59 Test whether the die is biased or not. [Given that the tabulated value of χ^2 at 5 % level of significance for 5 d. f. is 11.070.]	4	2	An	F

4	Show that the function defined by $f(z) = \sqrt{\lvert xy \rvert}$ satisfies Cauchy-Riemann equation at the origin but is not analytic at that point.	4	4	E	C
5	Prove that an analytic function with constant modulus is constant. **OR,** Prove that if $f(z) = u + iv$ is an analytic function then the family of curves $u(x, y) = a$ and $v(x, y) = b$ (where a & b are constants) cut each other orthogonally.	4	4	F	C

Section – B

*Attempt **All** Questions* $3 \times 5 = 15$ Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Construct an analytic function $f(z) = u + iv$ in terms of z whose real part is $u = e^x(x\cos y - y\sin y)$.	3	4	A	P
7	Evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz,$ where C is a circle $\lvert z \rvert = 3$.	3	4	E	C
8	Find the Laurent's series for the function $f(z) = \frac{1}{(z-1)(z-2)}$ in the following regions: (i) $1 < \lvert z \rvert < 2$ (ii) $\lvert z \rvert > 2$.	3	4	A	M

9	Show that the transformation $w = \frac{2z+3}{z-4}$ maps the circle $z\bar{z} - 2(z + \bar{z}) = 0$ into a straight line given by $2(w + \bar{w}) + 3 = 0.$ **OR,** Discuss the following types of singularities of a function f (z) with suitable examples: (a) Removable singularity (b) Essential Singularity (c) Pole	3	5	U	C
10	Find the residue of $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ at the singularities which lie inside the circle $\|z - 3\| = 1.5$.	3	4	E	P

Section – C

*Attempt **All** Questions.*

5 × 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Evaluate $\int_0^{2+i} z^2\, dz$ along: (a) the straight line joining $z = 0$ and $z = 2 + i$. (b) the real axis from $z = 0$ to $z = 2$ and then along a line parallel to imaginary axis from $z = 2$ to $z = 2 + i$. (c) the imaginary axis from $z = 0$ to $z = i$ and then along a line parallel to real axis from $z = i$ to $z = 2 + i$.	5	4	U	P

12	Find the bilinear transformation which maps the points $z = -1, 0, 1$ of the z-plane onto the points $w = -1, -i, 1$ of the w-plane. Show that under this transformation, the upper half of the z-plane maps onto the interior of the unit circle $\|w\| = 1$. **OR,** Define harmonic function. Show that $u = \frac{1}{2} log_e(x^2 + y^2)$ is a harmonic function. Also find its harmonic conjugate v such that $u + iv = f(z)$ is an analytic function of $z = x + iy$. Hence, determine the function $f(z)$ in terms of z.	5	5	A	P
13	Evaluate by the method of contour integration: $\int_0^{2\pi} \frac{1}{(2 + \cos\theta)} d\theta.$ **OR,** Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2 (z+2)}$ and the residues of $f(z)$ at each of its poles. Hence evaluate $\int_c f(z)\, dz$ where $c \equiv \|z\| = 3$ using Cauchy residue theorem.	5	6	An	C

Course Name: **English for Professional Purposes_BELH0003**

Course Outcome

CO1-The topics dealing with communication aspects will train the students hands on, by enabling them understand/identify and appreciate the barriers to communication, ways to overcome, process & type of communication etc.

CO2-The students will be able to understand the needs & requirements of the target audience based on their exposure and knowledge of the subject and will be able to communicate accordingly with certain level of accuracy

CO3-The writing skills of the students will be enhanced to such an extent that they could easily prepare and manage materials and strategies of formal correspondence themselves.

CO4-The students will be equipped with interpersonal skills through group presentations, skits and role-playing exercises. The exercises will help in getting rid of stage-fear, nervousness and hesitation during deliveries of official nature.

CO5-The students will be able to identify and rectify grammatical errors in written communication

Printed Pages: **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B.Tech (CS, EC, EE, ME, CE, Biotech), Second Year, III Semester
English for Professional Purposes I: BELH0003

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	The passage given below describes the role of teacher. It consists of 126 words. Write a précis and suggest a title. Teaching is one of the noblest professions. A teacher performs the sacred duty of making his students responsible, compassionate, and disciplined. Apart from developing their intellect, a teacher is also responsible for inculcating the qualities of good citizenship, cleanliness, politeness, and etiquette. These virtues are not easy to be imbibed. Only a person who himself leads a quality life characterized by simplicity, purity, and rigid discipline can successfully cultivate these habits in his pupils. A teacher always remains young at heart, although he may grow old age. Perpetual contact with budding youngsters keeps him hale and hearty. There are moments when domestic worries weigh heavily on his mind, but the delightful company of innocent children makes him overcome his transient moods of despair. (126 words)	4	3	R	F

2	Discuss flow of communication with diagram.	4	4	U	C
3	The CEO of Google, Mr. Sundar Pichai, is on a tour in India. He has agreed to pay a one-day visit to your University. The Dean of the University has asked you to carry out a survey amongst the students and then write a report. Write your report in about 200 words.	4	3	A	P
4	Give phrases, two each, useful in the group discussions while giving your opinion and while asking for an explanation.	4	2	R	F
5	Write a paragraph on the topic 'Cryptocurrency'. OR You are a part of the Corporate Communication team in your University. The working hours is revised as 9:30 AM to 5:30 PM due to cold temperatures in the winters. Using the following phrases, write an email with a minimum of 100 words and a maximum of 150 words to the employees in your company informing the same.	4	5	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
6	**i. Identify the type of construction mentioned within the bracket.** [*Whoever wants to go to Agra*] *may leave the class early* (a) Coordinate construction (b) Principal Clause (c) Relative Clause construction (d) Adverb Phrase (e) Interrogative sentence **ii. Specify the type of clause mentioned within the brackets:** *I want* [*to go to the university park*]. (a) independent clause (b) phrase (c) dependent clause (d) prepositional phrase (e) coordinate clause	3	5	A	C

	iii. Which combination makes a well-formed sentence in English? (a) Verb + Object (b) subject + Object/Complement (c) Subject + verb (d) Subject + verb + Object/Complement (e) Verb **In 4-6, choose the correct role for the highlighted verbs given in the sentence.** iv. Ravi will **get** the syllabus prepared by Shyam a) Passive verb b) Action verb c) Causative d) transitive e) intransitive v. Ram made his sister **buy** him a dress a) Action verb b) Causative c) Passive d) Transitive & Action e) Intransitive & Action vi. Shivam **brought** his wife berries a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb				
7	**Select the correct subject-verb combination in the given sentence (in 1-3):** i. One or two books are needed to complete the list. (a) Books and needed (b) One and needed (c) Two books and needed (d) Two books and are needed (e) One or two books and are needed. ii. Two thirds of the book is easy to read. (a) Two thirds of the book and to read (b) Book and is (c) Book and read (d) Thirds and is (e) None of the above	3	5	A	C

	iii. Neither Saloni nor her friends were going to attend the extra summer classes. (a) Saloni and were (b) Saloni and were going (c) Saloni, and to attend (d) his friends and were going (e) her friends and attend **In 4-6, choose the correct role for the highlighted verbs given in the sentence.** **iv.** The fictional characters will be **introduced** by Angira. a) Action verb b) Causative c) Passive d) Transitive & Action e) Intransitive & Action **v.** He made him **go** from the party. a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb **vi.** Sita **left** early in the morning. a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb				
8	**Choose the most suitable alternative to make the sentence logical in sense, from the given alternatives:** **i.** Recently, hopes that the World Bank will reach a deal to help India and Sri Lanka borrow at cheaper rates _____ nudged financial market higher. (a) are (b) has (c) have (d) will (e) **ii.** A great many players _____shown their interest in the exhibition match. (a) have (b)was (c) has (d) were (e) was **iii.**Two third of the pillar _____ rotten away. (a)has (b) have (c) are (d) were (e) is	3	5	A	C

	In 4-6, choose the correct role for the highlighted verbs given in the sentence. **iv.** I **have** him mix the jelly powder in the batter a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb **v.** Rajesh **got** the cake baked by his sister a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb **vi.** Reema **wants** to leave the college. a) Causative b) Passive c) Transitive & Action d) Intransitive & Action e) Action verb				
9	**Choose the most appropriate choice.** **i.** Riya has already spent the _____ money which I gave her last week. A. one B. some C. few D. all E. little **ii.** _____friend managed to complete it separately but together they completed it. A. Neither B. Both C. Some D. All E. Each **iii.** They had_____ rooms so none of them had any problems on accommodation. A. little B. few C. enough D. a E. the number of **iv.** These things happen in_____ home and ours is no different than____ A. any/the other B. every/others C. the others/the others D. each/ each other E. one/one another **v.** He was not very happy about it but he has _____ choice. A. neither B. both C. less D. little E. both C and D	3	5	A	C

	vi. The interviewer asked me_____ questions every _____minutes. A. several/many B. less/a few C. the number of/other D. many/ one another E. a number of/few				
10	**Choose the most appropriate choice from the given options.** i. The function started at ___noon and didn'tfinish until late in _____ evening. A. the/the B. no article/ no article C. no article/the D. the/an E. the/no article ii. Ramesh has gone to America for _____ months for his treatment. A. much more B. whole C. several D. plenty of E. both C and D iii. I liked_____ paintings, but not all of them were good. A. much B. few C. most D. many E. both C and D iv. She had _____ idea what the way to the theatre was. A. any B. more C. no D. none E. both B and C	3	5	A	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	**Specify the type of phrase and clause mentioned within the brackets.** i. He taught them science and history, and **played music for them.** A. Dependent clause B. Independent clause C. Noun phrase D. Verb phrase E. Adverb phrase ii.. **When we do not know our strength,** we search for	5	3	A	C

someone to help us.
A. Adjective phrase
B. Relative clause
C. Independent clause
D. Noun clause
E. Adverb Clause
A E

iii. An **economically backward** nation must invest in education.

A. Noun phrase
B. Adjective phrase
C. Adverb phrase
D. Verb phrase
E. Preposition phrase

iv. This **might have been done** by senior students.
A. Noun phrase
B. Adjective phrase
C. Adverb phrase
D. Verb phrase
E. Preposition phrase

v. There will be no class **on Saturday.**
A. Noun phrase
B. Adjective phrase
C. Adverb phrase
D. Verb phrase
E. Preposition phrase

vi. There will be no class **on Saturday.**
A. Noun phrase
B. Adjective phrase
C. Adverb phrase
D. Verb phrase

E. Preposition phrase

vii. The decade was a period of great prosperity **when industrial revolution took place.**

A. Noun clause
B. Relative clause
C. Adverb clause
D. Independent clause
E. Preposition phrase

Viii The minister announced **that the bridge would be constructed within a week.**
A. Noun phrase
B. Relative clause
C. Adverb clause
D. Subordinate clause
E. Preposition phrase

ix. I reached the college **before her examination had begun.**
A. Noun clause
B. Adverb clause
C. Independent clause
D. Relative clause
E. Preposition phrase

x. The book **on Dr. A P J Abdul Kalam** is very popular among the students.
A. Preposition phrase
B. Adjective phrase
C. Adverb phrase
D. Noun phrase
E. Relative clause

12	**Complete the sentences with an appropriate verb given.** i. One of the best students_____ _____ qualified this examination. A. is B. are C. has D. have E. has been ii. Measles _____ a common disease. A. are B. is C. were D. has E. does iii. Do you _____ these books? A. liking B. likes C. like D. have like E. liked iv. The courier _____ arrived. A. have B. are C. were D. has E. have been v. Either of the two candidates _____ eligible. A. is B. have C. are D. has been E. have been vi. They _____ in the tournament to represent his university. A. participates	5	3	A	C

B. was participating
C. participating
D. have participate
E. None of these

In 6-10, select the correct subject-verb combination in the following sentences-

vii. One or two projects are needed to complete the target.
A. One and needed
B. two projects and needed
C. two projects and are needed
D. one or two projects and are needed
E. one or two projects

viii. There was a series of Lockdowns in the country to curb the spread of Corona.
A. There and was
B. a series and was
C. a series of Lockdowns and curb
D. a series of Lockdowns and was
E. There and to curb

ix. Five hundred kilometer was not an enough distance.
A. Five hundred kilometer and was
B. an enough and was
C. kilometer and was
D. Five hundred and was
E. Distance and was

x. There is a box of cookies in the yellow car near the building.
A. There and is
B. A box and cookies
C. A box of cookies and is
D. There and car
E. There is and near the building

13	**i Choose the Noun Head in the sentence below.** *The people who are kind are always rewarded* (a) who (b) kind (c) rewarded (d) people (e) the people **ii. Choose the correct construction for B's reply** A: Where are the students? B: (a) garden (b) are in the garden (c) in the garden (d) the students in the garden (e) the garden **iii. Mark the independent/principal clause** *She did not understand whythe baby was crying* (a) why the baby was crying (b) the baby was crying (c) she did not understand why (d) she did not understand (e) why the baby **iv. What is the function of the relative clause in the sentence below.** *I went to the market which is far away from my residence* (a) modifier of the Noun *market* (b) post-modifier of the Noun *market* (c) adjective (d) Prepositional Phrase (e) interrogation **v. What kind of construction is the sentence illustrative of?** *Ram went to Agra, saw the Taj, and went to his cousin's place.* (a) Independent clause (b) Dependent clause (c) Coordinate clause (d) Verb Phrase (e) Simple sentence **vi. How many verbs are there in the sentence below?** *I would like to go to Delhi* (a) One (b) Two (c) Three (d) Four (e) Zero	5	3	A	C

	vii. How many meanings can you infer from the sentence? *The girl by the table with carved legs stood first in the class* (a) One (b) Two (c) Three (d) Four (e) Five **viii. Identify the Adjective Phrase in the sentence below** *Many of the exercises are fairly difficult* (a) Many (b) of the exercises (c) are fairly difficult (d) difficult (e) fairly difficult **ix. Find out the Adjective Phrase/s in the following sentence. Choose the correct option.** *She is very intelligent* (a) intelligent (b) very intelligent (c) she is (d) is intelligent (d) she **x. What are the constituents of a Noun Phrase? Choose the correct option.** (a) Noun + Verb (b) Preposition + determiner + Noun (c) determiner + Adverb + Adjective + Noun (d) determiner + verb + Adjective + Noun (e) Verb + Noun				

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 3 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B.Tech. (CSE), Year: II Semester: III
BCSC1003 Database Management System

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: ----------

1. Attempt all questions
2. Do not switch between the section A, Section B and Section C.
3. Use only blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the importance of data independence in the context of DBMS? Explain with suitable examples	4	1	A	C
2	Consider the following schema: Suppliers(*sid:* integer, sname: string, address: string) Parts(*pid:* integer, pname: string, color: string) Catalog(*sid:* integer, *pid:* integer, cost: float) Write the following queries in relational algebra: a. Find the *names of suppliers* who live in "Patna." b. Find the *details* of parts whose color is red. c. Find the *sids* of the supplier who supplies part having cost greater than 10. d. Find the *sids* of suppliers who supply some red parts.	4	3	A	M

3	Differentiate between the following: a. DDL and DML b. Natural Join and Theta Join	4	2	U	C
4	Discuss ACID properties and explain each property with help of example.	4	7	R	F
5	What is Distributed database? Discuss fragments and its type with help of example?	4	8	U	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Define Boyce-Codd Normal Form. How does it differ from 3NF?	3	7	U	C
7	Consider the following Relation $R(ABCDEFGHIJ)$ with following set of FDs $F = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$ Answer the following: a. Find all the candidate keys? b. Find the Highest Normal Form. c. Decompose R upto $3NF$	3	6	E	M
8	What are the various states of transactions, can you arrange these states of transactions in a sequence, if yes, represent the sequence in a diagram.	3	7	A	P
9	Discuss two-phase locking protocol. What is the limitation of 2PL?	3	8	U	F
10	Consider a relation schema R (A, B, C, D) and functional dependencies A → B and C → D. State whether the following decomposition of relation R into R1 (A, B) and R2(C, D) is the Loss Less Join Decomposition or Lossy.	3	6	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	What is Log? Discuss the salient features of Deferred database modification and Immediate database modification strategies in brief.	5	5	U	F
12	Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. (a) $r_1(X)$; $r_3(X)$; $w_1(X)$; $r_2(X)$; $w_3(X)$ (b) $r_1(X)$; $r_3(X)$; $w_3(X)$; $w_1(X)$; $r_2(X)$ (c) $r_3(X)$; $r_2(X)$; $w_3(X)$; $r_1(X)$; $w_1(X)$ (d) $r_3(X)$; $r_2(X)$; $r_1(X)$; $w_3(X)$; $w_1(X)$	5	8	C	M
13	What is Indexing in DBMS? Explain dense and sparse indexing with an example?	5	7	U	F

Course Name: Introduction to Machine Learning

Course Outcome

CO1- Apply the basic concepts of machine learning including bias-variance tradeoff.
CO2- Apply the concepts of regression.
CO3- Conceptualize supervise and re-enforcement learning for classification.
CO4- Formulate the ensemble methods for improving classification.
CO5- Apply ANN with optimization in machine learning

Printed Pages: 3 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B.Tech (CSE – AIML), Year: II, Semester: II
BCSE 0701, Introduction to Machine Learning

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: ----------

- *This paper is divided into three sections: **A, B and C**. All the sections are **compulsory**.*
- *Write down the Serial Number of the question before attempting it and do all questions of a section at one place.*

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is 'Overfitting' in Machine learning? Why does overfitting happen?	4	1	U	C
2	What are the different Algorithm techniques in Machine Learning?	4	1	R	F
3	An art competition has entries for three painters: Vipin, Rohit and Ajay. • Vipin put in 15 paintings, 4% of his works have won First Prize. • Rohit put in 5 paintings, 6% of his works have won First Prize. • Ajay put in 10 paintings, 3% of his works have won First Prize. What is the chance that Vipin will win First Prize?	4	3	A	P

4	Given the confusion matrix, find: Classification Accuracy, Recall, Precision, F-measure. N=165 | Predicted: No | Predicted: Yes | Actual: No | Tn=50 | Fp=10 | 60 Actual: Yes | Fn=5 | Tp=100 | 105 55 | 110	4	3	A	P
5	What are Support Vectors in SVM? What are Different Kernels in SVM?	4	3	R	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Write down the difference between feature selection and feature extraction.	3	1	U	C
7	What is the dimensionality reduction? How is it achieved in Machine Learning?	3	1	A	C
8	What is NLTK? How is it different from Spacy?	3	3	R	F
9	How Agglomerative Hierarchical clustering algorithm works?	3	4	R	F
10	Explain the differences between ANN and CNN. Write down the python libraries to implement CNN.	3	4	U	C

Section – C

*Attempt **All** Questions*

5 X 3 = 15 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>11</td><td>What are ensemble methods? Differentiate between bagging and boosting ensemble methods</td><td>5</td><td>4</td><td>R</td><td>F</td></tr>
<tr><td>12</td><td>Apply the ID3 algorithm for decision tree learning in the following dataset. One can find a particular type of food is appealing based on the food's temperature, taste, and size.

<table><tr><th>Appealing</th><th>Temperature</th><th>Taste</th><th>Size</th></tr><tr><td>No</td><td>Hot</td><td>Salty</td><td>Small</td></tr><tr><td>No</td><td>Cold</td><td>Sweet</td><td>Large</td></tr><tr><td>No</td><td>Cold</td><td>Sweet</td><td>Large</td></tr><tr><td>Yes</td><td>Cold</td><td>Sour</td><td>Small</td></tr><tr><td>Yes</td><td>H</td><td>Sour</td><td>Small</td></tr><tr><td>No</td><td>H</td><td>Salty</td><td>Large</td></tr><tr><td>Yes</td><td>H</td><td>Sour</td><td>Large</td></tr><tr><td>Yes</td><td>Cold</td><td>Sweet</td><td>Small</td></tr><tr><td>Yes</td><td>Cold</td><td>Sweet</td><td>Small</td></tr><tr><td>No</td><td>H</td><td>Salty</td><td>Large</td></tr></table>
Which attribute will ID3 place at the root? Explain the process.</td><td>5</td><td>3</td><td>A</td><td>P</td></tr>
<tr><td>13</td><td>Cluster the following eight points (with (x, y) representing locations) into three clusters A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2) and A8(4, 9).

Initial cluster centers are A1(2, 10), A4(5, 8) and A7(1, 2). Use k-means algorithm to find the three cluster centers after the second iteration.</td><td>5</td><td>1</td><td>A</td><td>P</td></tr>
</table>

Course Name: Introduction to Virtualization & Cloud Computing

Course Outcome:

CO1: Understand the basics of virtualization technology and hypervisors.
CO2: Analyze the trade-offs between deploying applications in the cloud and over the local infrastructure.
CO3: Recognize real-world problem using cloud computing through group collaboration.
CO4: Understand on-demand utility computing phenomenon of cloud computing.
CO5: Understand the issues involved in cloud computing.

Printed Pages: 02 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B.Tech CCV, II Year, III Semester
BCSE 1502 Introduction to Virtualization & Cloud Computing

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Any request you made to a server over internet is transferred in plain text form. Suggest a network security technique that provides various security attributes and also help to hide your identity. Compare the suggested technique with network proxy.	4	2	An	M
2	Identify the types of virtualization, that an organization should use, who is planning to set up its various offices in different part of the world. Justify your answer with proper reasoning. Also, discuss the cost impacts of proposed virtualization.	4	1	E	P
3	Virtual servers run on a hypervisor, and hardware resources are emulated for the virtual servers via the hypervisors. If the hypervisor fails or if the underlying physical server fails (thereby causing the hypervisor to fail), the failure condition cascades to all of its hosted virtual servers. Design a cloud architecture to overcome this failure.	4	1	An	M
4	Based on Total Cost of Ownership and Return on Investment, out of public and private cloud which will be more economical to end user. Elaborate your answer.	4	3	A	C
5	A school having nice computer lab facility wants to go on cloud so as to decrease the cost of licensing software. The infrastructure that the school is having is updated and is in good condition.	4	3	An	M

	However, they are reluctant to move on cloud, thinking it to be expensive. Neither they want their database to move out of premises. Examine the requirements of the school and identify the deployment model and cloud services. Also, justify your answer.				

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Which cloud model solves the issue of vendor lock-in? How the suggested model is different from multi- cloud architecture? Also, define the term cloud bursting.	3	4	R	C
7	Differentiate between: **a.** Full virtualization and Para virtualization. **b.** Cluster and Grid Computing.	3	4	An	M
8	Differentiate between cloud computing and virtualization.	3	3	An	M
9	Explain anatomy of cloud with the help of example.	3	5	U	F
10	Explain resource pooling and multi- tenancy with the help of example.	3	4	R	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	An organization want a storage solution with no single point of failure. Identify a storage solution that match the requirement of the organization, also mention other features of the storage solution in support of your answer. Discuss about various RAID levels with their pros and cons.	5	3	An	M
12	Define cloud deployment decision factors with the help of example.	5	4	R	C
13	Design a factor matrix for public cloud based on Business IT Control, CAPEX & OPEX, Workload characteristics, workload lifespan, industry segment and data freedom.	5	5	U	P

Course Name: IT Application and Data Security

Course Outcome

CO1-Understand software development methodology and application security

CO2-: Explain input validation strategies; protect sensitive data, effective authentication and authorization strategies.

CO3- Understand various attacks like DoS, buffer overflow, web specific and database specific.

CO4- Implement security as a culture to show mistakes that make applications vulnerable to attacks.

CO5- Demonstrate the security enhancements of web based applications.

CO6- Understand the basics of data security threats.

CO7- Apply the data security threat techniques and its counter measures

Printed Pages:2 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
Program: B. TECH(CSF), Year: II, Semester: III
Subject Code & Name- IT Application and Data Security (BCSE 0601)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Define threat and malware? How the phases used in the traditional Software Development Life Cycle can be adapted to support the implementation of an information security project. Give a detailed analysis.	4	1	R	F
2	Discuss the most effective countermeasures of input validation and write down the practical solutions for input validation. Also explain the different ways to prevent data tampering.	4	2	U	C
3	Differentiate between Authentication and Authorization. Explain session hijacking, session replay and Man in Middle attack with example.	4	2	An	F
4	What do you understand by data security? Discuss and explain the lifecycle of data security with suitable diagram.	4	6	R	F
5	Write short note on the followings: A. Banking Frauds Counter Measures B. Blue- bugging and Blue- smack C. Symmetric and Asymmetric Cryptography D. Intrusion Detection Systems(IDS)	4	7	A	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Discuss and explain the technique where values and actions are inserted unexpectedly as an application's input to crash it down.	3	3	U	
7	The academic security model was "wide open" and the government security model was "closed and locked". Justify this statement with suitable example.	3	7	E	P
8	How Auditing and logging can be used for detecting the suspicious activities? Discuss the various threats related to auditing and logging.	3	4	U	C
9	Suppose your computer's CPU usage is more than 50%, and outgoing network activity is far from normal and you suspect that your computer may be part of a botnet. How will you ensure this? Explain the various preventive measures for the same.	3	4	U	P
10	Differentiate between Birthday attack and cryptanalysis attack. Write down the various steps required to perform SQL Injection.	3	7	An	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Discuss and explain the process in which an attacker covers tracks by destroying all evidence of the attack. Also explain four strategic countermeasures for protecting Web applications and to increase security awareness.	5	7	U	F
12	Which types of sniffing attacks can be carried out at every layer of OSI-Model? Also discuss counter measures for cross-site scripting forgery.	5	6	R	C
13	A. What do you understand by Centralized Key Management? Explain Components of Centralized Key Management with suitable example. B. Discuss the various Hacking & Social Networking Countermeasures	5	5	R	C

Course Name: Data Visualization & EDA

Course Outcome

CO1: Apply the basic concepts of Data Visualization.
CO2: List various business moment decisions.
CO3: Apply the concept of data preprocessing.
CO4: Differentiate Data types .
CO5: Implement Data Preprocessing technique.

Printed Pages: 02 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B. Tech (H.), II Year, III Semester
BCSS 0051 : Data Visualization & EDA

Time: 3 Hours **Maximum Marks: 45**

Instruction for students: Write answer of a section at one place. Provide necessary code wherever required. Take care of indentation properly.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Define Data Visualization. What makes a good data visualization.	4	1	R,U	C
2	What is a scatter plot? What types of data work best in scatter plots?	4	2	U	C
3	When would you use a pie chart vs a line chart? Propose examples for each.	4	1,2	AN	F
4	What is a heat map in Python? Create a correlation matrix using the core function of the data frame?	4	4	C	PC
5	What would you do if you suspected that your data was inaccurate?	4	3	AN	C

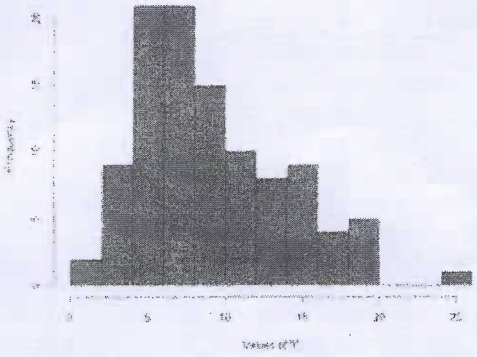
Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain when you might use a bar graph over a pie chart?	3	1,2	U	C
7	What are some common issues you encounter when doing data visualization with large amounts of data?	3	3	R	F
8	Write a program in python to show the map using markers	3	1	C	PC
9	Write a note on univariate, bivariate and multivariate data visualization	3	1	R	F
10	Differentiate the following: i) Bar plot and count plot ii) Regression plot and line plot	3	2	A	C

Section – C

Attempt All Questions 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Answer the following three questions based on the histogram above. (i) Where would the mode of this dataset lie? (ii) Comment on the skewness of the dataset. (iii) Suppose that the above histogram and the boxplot in question 2 are plotted for the same dataset. Explain how these graphs complement 	5	2	E	C
12	Below are the scores obtained by a student in tests 34,36,36,38,38,39,39,40,40,41,41,41,41,42,42,45,49,56 1) Find mean, median, variance, standard deviation. 2) What can we say about the student marks? List out some graph and chart that might be useful to show these result and display these graph using plotly library in python.	5	1,2	E	C

Format of Question Paper for End- Term Examination (Semester System)

Course Name: BCSE0551: INTRODUCTION TO BUSINESS ANALYTICS

Course Outcome

CO1: Understand the use of reference architecture in business analytics.

CO2: Explain the decision support science and data warehouse.

CO3: Understand the concepts of Business Intelligence.

CO4: Explain the KDD in data mining.

CO5: Understand the process of analysis using Dashboards and Reports.

CO6: Explain Big Data, Filtering Big Data and Adoption Architecture.

Printed Pages: 03 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23

B.Tech. (CSE - DA), II, III Sem

BCSE0551: INTRODUCTION TO BUSINESS ANALYTICS

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions
2. do not switch between section A, B and Section C; first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Design a bottom-up architecture of Data Ware House for any health insurance sector. Your stepwise design process must have proper explanation.	4	2	C	P
2	Discuss ROLAP, MOLAP and HOLAP. Differentiate between OLTP and OLAP. What are the features of data warehouse?	4	2	An	C
3	Discuss BI Architecture. What are the different stages and steps involving in Bi Design and development?	4	3	E	C
4	What is Machine Learning? Discuss Supervised Learning, Unsupervised Learning and Reinforcement Learning with help of example.	4	6	U	P

5	What is data mining? What are the challenges of Data mining? Is Big Data proving to be useful in overcoming any of these challenges?	4	4	E	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Discuss how business analytics can be used in sports, such as tennis, cricket, football and so on. Identify as many opportunities as you can for each.	3	1	C	C
7	Suggest some metrics that a railway or bus ticketing agency might want to collect. Describe how a manager might utilize this data to facilitate better decisions.	3	6	E	P
8	Elaborate on functional pillars and foundation layers of BAO.	3	1	U	P
9	Write a note on the importance of Process standardization and CRISP-DM.	3	4	R	P
10	As per IBM Think Tanks CAMSS (Cloud, Analytics, Mobile, Social & Security) is future for IT companies, (a) As a BAO domain expert, what role you define for yourself. (b) Draw the importance of BAO in Cloud, Big Data, Artificial Intelligence, Machine Learning, NLP (Neural Language Processing), HealthCare Image Processing. (c) Write IBM Applications or Tools used for three category of Business Analytics.	3	5	A	M

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Answer the following: (a) Discuss layers of information. (b) What are the different Gestalt principles of perception applicable to dashboards?	5	5	U	P
12	Discuss following with suitable diagram: a. Line chart b. Pie chart c. Radar chart d. Column chart e. Bar chart f. Gauges	5	6	An	P
13	What are the three types of dashboard? Explain them vividly.	5	5	U	P

Course Name: B.Tech (Hons.)

Course Outcome

CO1- Understand the concept of database management systems and Relational database..

CO2- Identify the various data model used in database design.

CO3- Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.

CO4- Create and populate a RDBMS for a real life application, with constraints and keys, using SQL.

CO5- Discuss recovery system and be familiar with introduction to web database, distributed databases.

CO6- Select the information from a database by formulating complex queries in SQL.

Printed Pages: **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
Program: B. Tech (H), Year: II, Semester:III
Subject Code & Subject Name- BCSC0054 Advanced Database Management System

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following tables: Customer(Cutomer_Name, street, city) Branch(branch_name, branch_city) Account(acc_no, branch_name, balance) Loan(Loan_no, branch_name, amount) Borrower(Cutomer_Name, Loan_no) Depositor(Cutomer_Name, acc_no) Write the relational algebra expression for the following queries: a. Find the loan number, branch, amount of loans of greater than or equal to 10000 amount. b. Find the loan number for each loan of an amount greater or equal to 10000 c. Find the names of all customers who have a loan and an account at the bank. d. Find the names of all customers having a loan at the "ABC" branch.	4	1	R	F
2	Differentiate between serial schedules and serializable schedules. Classify serializability with proper example.	4	2	A	C

3	Define concurrency in transaction. What problems could arise due to concurrency in transaction?	4	1	U	C
4	Check whether the given schedule S is view serializable or not. If yes, then give the serial schedule. S : $R_1(A)$, $W_2(A)$, $R_3(A)$,-$W_1(A)$, $W_3(A)$	4	3	R	C
5	What is transaction and state its various states, explain with suitable diagram?	4	1	U	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following relation schema: College (Cname, State, enrollment) Student (Sid, Sname, GPA, Sizehs, Dob) Apply (Sid, Cname, Major, Decision) Write the Sql queries for the following : **a)** Find detail of applications who applied to California State. **b)** Find Sids, name, GPA of students and name of college with GPA > 3.7 applying to Stanford. **c)** Find details of students who have not applied to any of college.	3	4	R	F
7	Consider the relation schema given in question 6, Write the sql queries for the following: **a)** Find how many applications were rejected and accepted by the colleges. **b)** Find number of applications received by particular college. **c)** Find how many different majors student had applied.	3	4	U	F
8	What is referential integrity constraint, explain with suitable sql query?	3	3	R	F
9	Consider the following relation: EMP_PROJ(Emp#, Proj#, Ename, Pname, No_hours). (Consider Emp# and Proj# as key attribute)	3	5	U	F

	Explain which type of anomaly is in the given relation? And how would you remove it?				
10	Consider the following relation given below, find the highest normal form of the relation, and normalize it further. **DEPARTMENT** Dname | Dnumber | Dmgr_ssn | Dlocations Research | 5 | 333445555 | {Bellaire, Sugarland, Houston} Administration | 4 | 987654321 | {Stafford} Headquarters | 1 | 888665555 | {Houston}	3	2	R	F

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider the following set of operations in the log. $(start, T4)$ $(write, T4, y, 2, 3)$ $(start, T1)$ $(commit, T4)$ $(write, T1, z, 5, 7)$ $(checkpoint)$ $(start, T2)$ $(write, T2, x, 1, 9)$ $(commit, T2)$ $(start, T3)$ $(write, T3, z, 7, 2)$ If a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list.	5	3	A	P
12	What are the different two phase locking protocol? Explain each of them with their pros and cons.	5	4	A	P

Course Name: Probability &Applied Statistics

Course Outcome

CO1: List different kind of statistics for data analysis.

CO2: Differentiate measure of central tendency and measure of variability.

CO3: Define Normalization and Standardization.

CO4: Conceptualize Probability distribution

CO5: Apply statistics in various research methods

Printed Pages: 03 **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B. Tech (H.), II Year, III Semester
BCSS 0052 : Probability &Applied Statistics

Time: 3 Hours Maximum Marks: 45

Instruction for students: Write answer of a section at one place. Provide necessary code wherever required. Take care of indentation properly.

Section – A

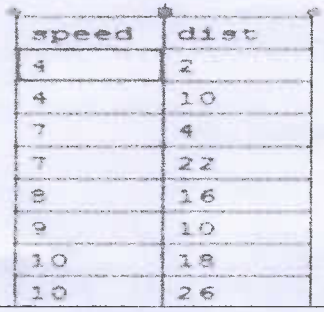
Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is Hypothesis Testing and when do we use it?	4	1	R	F
2	Draw the flow chart and explain all the step we performs in Anova testing.	4	2	AN	D
3	Define correlation. Write a program to find out the correlation between the different variable by i) Statistical value ii) Using a suitable graph Use data as per your suitability by creating a numpy array.	4	1,2	R,C	F,P
4	Can you explain a possible method of improving the accuracy of a linear regression model?	4	4	AN	C
5	i) If you have only one independent variable, how many coefficients will you require to estimate in a simple linear regression model? ii) What is the primary difference between R square and adjusted R square?	4	3	R,C	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	i) What is Heteroscedasticity? ii) What is Homoscedasticity? iii) What is p-value	3	1,2	R	F

7	i) Differentiate between regression and classification problem. ii) Differentiate Simple linear regression and Multiple linear regression	3	3	AN	C
8	Suppose following dataset is given . Write a program in python to build a simple linear regression model that will predict the speed on	3	5	A	PC
	the basis of the distance given. 				
9	Define the followings: i) VIF value ii) Influential observation iii) Cook's Distance	3	1	R	F
10	i) You have a cleaned data set and you want to perform analysis using statistical models. Which method can you use to show the facts as well as an interpretation of what those facts indicate in the context of your project? ii) You are a data analyst, and you are performing the cleaning and organized data activity. What is the rule to be followed to show the percentage of time dedicated to cleaning and organizing data compared to analysis?	3	5	AN	C

speed	dist
4	2
4	10
7	4
7	22
8	16
9	10
10	18
10	26

Section – C

Attempt All Questions 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Assume we are pizza makers and we are interested to check if the diameter of the Pizza follows a Normal/Gaussian distribution? Dataset—diameter.csv i) Write the null and alternate hypothesis for above problem ii) Write a program in python to do this hypothesis testing	5	1.5	AN	PC
2	Telecall uses 4 centers around the globe to process customer order forms. They audit a certain % of the customer order forms. Any error in order form renders it defective and must be reworked before processing. The manager wants to check whether the defective % varies by center. Please analyze the data at 5% significance level and help the manager draw appropriate inferences. i) Write a flow chart to do whatever hypothesis testing is required ii) List all the function that may be required to perform each step mentioned in the flow chart	5	1.5	AN	PC

Course Name: Ethics & Values

Course Outcome

CO1-Development of Rationality, Critical Thinking and Problem Solving.
CO2-Profound Ethical Decision Making and Value Judgment Capacity.
CO3-The enhanced ability to understand the ethical problems of personal, social & professional life.
CO4-Well constructed professional approach to rationalize life issues through developed life-skills.
CO5-Understanding of Personal, Professional & Social ethics.

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End Term Examination, ODD Semester 2021-22
B. Tech. (Hons) II Year, III Semester
BELH-0017 Ethics & Values

Time: 2 Hours **Maximum Marks: 45**

Instruction for students: Attempt all questions.

Section – A **Objective Type** **.5 X 25 = 12.5 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	What are the similarities between the case of cannibalism and the run way trolley? A. Rationality B. Critical thinking C. EDM D. All of these	1/2	5	U	C
2	Why does argument from necessity disapprove? A. Rescue was sure B. Rescue did not reach C. Necessity was not there D. None of these	1/2	1	E	F
3	What kind of necessity was there in the case of cannibalism: A. Necessity of food B. Necessity of survival C. Necessity of killing Parker D. None of these	1/2	2	A	, P
4	Which were the real arguments? A. Argument from intention B. argument from lottery C. Argument from necessity D. Both A & C	1/2	1	AN	M
5	Who killed Parker? A. Brooke B. Dudley C. Stephenson D. All of these.	1/2	3	C,	, P
6	The case of cannibalism involves the following perspective: A. Profit B. Utilitarianism C. Hypothetical D. None of these	1/2	2	U	P
7	Student should be disciplined: A. Moral B. Necessary C. Both D. Both Not	1/2	3	An	R

8	A professional should care profession: A. Necessary B. Moral C. Both D. Both Not	1/2	4	E	R
9	Story of stuff does not discuss: A. Environmental Ethics B. capacity to think C. capacity to cheat D. both b&c	1/2	5	AN	M
10	Our action should be: A. Environment friendly B. Producing carbon C. Harming nature D. None of these	1/2	4	C	C
11	Modern times' central theme is: A. Keep trying B. Keep Idle C. Creating problems D. None of these	1/2	5	U	C
12	What is depicted in the documentary: Modern Times? A. Mechanical system B. Ethical System C. Inhuman System D. Both a&c	1/2	1	E	F
13	Charlie got mad in a scene because of: A.Mechanical system B. ethical system C. good system D. All of these	1/2	2	A	, P
14	The Modern Times is a criticism of : A. Mechanical World B. ethical world C. Money D. Both a & b	1/2	1	AN	M
15	Charlie helped Jailor because: A. He was a moral person B. He helped for himself C. Charlie was drunk D. none of these	1/2	3	C,	, P
16	Right Here and Right Now teaches : A. Maintain harmony B. Negative attitude is good C. Positive attitude is bad D. All of these	1/2	2	U	P
17	Elder brother misbehaved with mother led : A.Negative B. maid left the job C. accident D. All of these	1/2	3	An	R
18	Boss drove the car and dropped the driver because : A. He was happy B. happiness leads valuable behavior C. had nothing to buy D. none of these	1/2	4	E	R

19	In the last scene Doctor saved the person but could not kept the promise: A. He understood human values B. professional ethics C. self interest D. all of these	1/2	5	AN	M
20	What is sustainability: A. Saving for the next generation B. Producing carbon C. Harming nature D. None of these	1/2	4	C	C
21	Who talks about the distributive justice: A. John Rawls B. I. Kant C. J. S. Mill D. J. Bentham	1/2	5	AN	M
22	Libertarianism is based on: A. Utility B. Duty C. Individual Rights D. All of these	1/2	4	C	C
23	Ethics is about: A. Only good B. Only Right C. Only duty D. None of these	1/2	4	C	C
24	Who wrote On Liberty: A. John Rawls B. Kant C. Bentham D. None of these	1/2	5	AN	M
25	Justice is attributive: A. Aristotle B. Mill C. Bentham D. All of these	1/	4	C	C

Section – B **Multiple Choice** **.5 X 25 = 12.5 Marks**

No.	Detail of Question	Marks	CO	BL	KL
26	Taxation is unfair according to Liberals: 1. Violates freedom 2. Ownership 3. Forced Labor 4. None of these A. Only 1 B. 2& 4 C. 1,2& 3 D. 4	1/2	4	U,	M
27	What are the objection by Liberals? 1. Moral Legislation 2. No Taxation 3. Paternalistic Laws4. No Objection A. 1, 2, & 3 B. only 1 C. Only 2 D. Only 3& 4	1/2	1	An,	F
28	Minimal state theory maintains: 1. To maintain liberty 2. state should not excessively intervene 3. Individual freedom 4. none of these A. 2& 3, 4 B. 1,2, 4 C . 1,2,3 D. 1& 4	1/2	4	A	P

29	Freedom Means: 1. Complete 2. But Not Unlimited 3. Only as means 4. Respect for powers A. 4 & 2 B. 1,2 C. Only 1,3 D. Only 4	1/2	5	A	C
30	Libertarians objects moral legislation: 1. Individual should be free to make choices 2. state should not impose choices 3. for immorality 4. For money A. 1& 2 B. 2& 3 C. 1,2,3,4 D. Only 1& 4	1/2	1	An	R
31.	Taxation is wrong: 1. Profession becomes slavery2. It's not moral 3. Job becomes forced labour 4. State shares ownership A. 1,2,3,4 B. 2 & 4 C. 1& 3 D. Only 4	1/2	2	U	C
32.	Paternalistic laws are : 1. Driving safety 2. Pension 3. Medical Insurance4. Bearing helmate A. 1,2,3, 4 B. 1,2 C. 2& 4 D. 3 & 4	1/2	3	E	M
33	What is affirmative action: 1. Benifiting powerul 2. Dominating weak 3. Benefiting creamy layer4. None of these A. 1,2 B. 1,3 C. 1,2,3, D. only 4	1/2	3	E	F
34	What are the three arguments for Affirmative Action: 1. Corrective 2. Interrogative 3. Compensatory 4. Social Representation A. 1,2 B. 2,3 C. 3,4 D. 1,3,4	1/2	2	An	R
35	Corrective argument put emphasis on: 1. Future wrong 2. Correcting justice 3. Correcting right 4. None of these A. 1,2 B. only 4 C. 3,4 D. 2,3,4	1/2	5	C	P
36.	What are the similarities between the corrective and compensatory arguments: 1.correcting the future 2. For reservation 3. For social justice4. Social Representation A.1,2 B. 1,2,3 C. 2,3D. 1,3,4	1/2	2	An	R
37	Social Representation argues for: 1.Representation of people 2. Affirmative Action 3. Social Justice 4. Social representation A.1,2,3,4 B. 2,4 C. 1,2,3 D. 2,3,4	1/2	5	C	P

38	Deontology does not focuse on : 1. Duty 2. Money 3. Freedom 4. Result A.1,2 B. 2,4 C. 1,3,4 D. 2,3,4	1/2	5	C	P
39.	Human being differs from objects & animals: 1. They are free2.They can reason3. They can choose 4. None of these A.1,2,3 B. 2,3 C. 3,4 D. 1,3,4	1/2	2	An	R
40	Morality includes: 1. Opinion 2. individuality 3. subjectivity 4. None of these A.1,2,4 B. 2,4 C. 1,2,3 D. 2,3,4	1/2	5	C	P
41	Principle of eqiality means: 1. Equal rights 2. Equal Liberties 3. Experiment by Rawls 4. None of these A.1,2,4 B. 1,2 C. 1,3,4 D. 2,3,4	1/2	5	C	P
42.	Justice as fairness 1. Rawls talks on it 2. fair systems3. fair distribution 4. gender A.1,2,3 B. 2,3 C. 3,4 D. 1,4.	1/2	2	An	R
43	What is moral luck: 1. Fairness 2. Distributive 3. Good by chance 4. Moral credit A.1,2, B. 2,4 C. 1,2,3 D. only 3	1/2	5	C	P
44	Who does not deserve the best flute: 1. By interest 2. by merit 3. by money 4. None of these A.Only 1,2 B. 2,4 C. 2,3 D. 2,3,4	1/2	5	C	P
45	Kidney selling in market would not be wrong by Liberals because it is our : 1.Ownership 2. Freedom 3. power 4. None of these A.1,2 B. 1,2,3 C. 1,3,4 D. 2,3,4	1/2	5	C	P
46	Respect is by Liberals: 1.As Ends 2. Nothing 3. Complete 4. Not as means A.1,2 B. 1,2,3 C. 3,4 D. 1,4	1/2	2	An	R
47	What are two principles of justice: 1.Principle of Equality 2. Principle of Difference 3. Principle of Utility 4. All of these A. 1,2, B. 2,4 C. 1,2,3 D. 2,3,4	1/2	5	C	P
48	Which kind of surrogacy is not permissible in India : 1. Philanthropic 2. Paid 3. In relationship 4. Comercial A. 1,2 B. 1,2,3 C. 1,4 D. 2,3,4	1/2	5	C	P

49	What is Active Euthanasia: 1. Suicide 2. mercy killing 3. assisted killing by state 4. none of these A.1,2 B. 2,3 C. 3,4 D. 1,3	1/2	2	An	R
50	Passive mercy killing allows: 1. can stop medication 2. Active 3. get discarge from hospital 4. All of these A.1 B. 1,3 C. 1,2,3 D. 2,3,4	1/2	5	C	P

Section – C Assertion (A) and Reason (R) Based **1X 20 = 20 Marks**

Fill the bubbles as per following instructions for this section:

A: If only Assertion is Correct.

B: If only Reason is Correct.

C: Both Assertion and Reason are correct and Reason is right for the Assertion.

D. Both Assertion and Reason are correct but Reason is not right for the Assertion.

No.	Detail of Question	Marks	CO	BL	KL
51	A: I am ethical. R: I follow rationality.	1	1	An	M
52	A:The argument from necessity fails. R: Necessity for survival was there.	1	2	An	C,
53	A: Exam grades matter. R: Grades show your performance.	1	3	U	C
54	A: Ethical Decision Making is descriptive ethics . R: EDM involves analysis of norms, decisions and policies .	1	4	E	F
55	A: Justice is attributive. R : It depends on fair distribution.	1	5	C	C
56	A: Moral Desert is ethical principle. R: One should be favored not by virtues .	1	3	U	F
57	A: Humanity principle treats human as ends. R : Treat humans always as means.	1	1	E	R,
58	A: I can make moral rules. R: My rule can be a moral law if it can be universally applied.	1	4	An,	M,

59	A: Morality requires freedom. R: With freedom I cannot be held moral.	1	2	E	C
60	A: According to liberals freedom is incomplete R: I can even harm myself.	1	1	An	M
61	A:Mercy Killing is suicide. R: In mercy killing one is unable to live.	1	1	An	M
62	A:In active euthanasia state provides aid to die. R: India allows passive mercy killing.	1	2	An	C,
63	A: Liberals defend individual rights. R: Individual rights are necessery for the moral life.	1	3	U	C
64	A: Taxation is unfair . R: It is a violation of ownership .	1	4	E	F
65	A:Taxation is fair. R :Redistribution of wealth is to maintain social justice.	1	5	C	C
66	A: Best players deserve flutes. R:They have interets in flute-playing.	1	3	U	F
67	A: Lectures are meant for knowledge & learning. R : This is the purpose of academics.	1	1	E	R,
68	A: I study hard to get the good marks . R: This is the hypothetical reason.	1	4	An,	M,
69	A: Surrogacy is a bioethical issue. R: If it is done for philanthropic reasons;it's moral.	1	2	E	C
70	A: India allows surrogacy. R: Only philanthropic surrogacy is legal in India.	1	1	An	M

Course Name: ENGLISH FOR OCCUPATIONAL PURPOSES- I

Course Outcome

CO1- Understand the basic grammatical skills
CO2- Apply the prescribed language skills in writing
CO3- Express ideas in writing according to time and tense
CO4- Analyze the comprehension and answer the questions based on it
CO5- Enrich vocabulary in terms of contextual and situational conversation

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End Term Examination, Odd Semester 2022-23

B. Tech. (Hons.) CS, BCA (Hons.), II Year, III Semester

BELA 0001: ENGLISH FOR OCCUPATIONAL PURPOSES- I

Time: 3 Hours **Maximum Marks:45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Describe the site of alien spaceship which you viewed from the terrace of your house during pandemic in 150 words.	4	3	AN	F
2	A) Do as Directed: i) The interest rate in Brazil fell from 25% in 2003 to 11.25% in 2006. (Identify the prepositional phrase) ii) Make two separate sentences using the following adjectives: 'luxurious' and 'preventive' iii) Use the following words as adjective and adverb in two separate sentence: high, like iv)In some disciplines, women equal or (even) outnumber their male colleagues. (Identify the verb phrase) Or A) Do as Directed: i) Globalisation has brought enormous opportunities for investment but it has exposed serious risks.(Identify the principal clause and subordinate clause) ii) Identify the base word and three morphemes from which the word "globalisation" is driven. iii) So that they could reduce costs and increase productivity, chemists invented chlorofluorocarbons. (Identify the sentence on the basis of its structure) iv) Those biology classes we attended last week. (Identify the pre- modifier and post modifier) v) Several policies have been set by the government, at all levels, to assist low income households. (Underline the noun group-complexes)	4	1	A	C
3	You are being called for an interview for the post of Web Developer at Central Business Solutions Inc, India. The interview panel has asked you to introduce yourself. Write your introduction giving details about your background, educational qualifications, extra and co-curricular activities, achievements and interests.	4	2	R	P

4	Write a report on the Mini Project on "Sales forecasting System for E- Commerce food products" which you recently completed.	4	3	A	P
5	Do as directed: i) Name the word formation process/s of the following words: Edutainment, breakdown, mart ii) Form words combining the given word elements and use them in sentences: Net + citizen, Hack+ marathon iii) Find suffixes in these words and make two new words from them: Regulate, boredom	4	3	AN	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Do as directed: i) Making education cheaper will help **alleviate** poverty. (Use antonym of the underlined word) ii) The organization needs to **leverage** its key resources. ((Use synonym of the underlined word) iii) Metal and mettle (Make sentences to differentiate the meaning) iv) Use the following polysemous words to indicate difference in meaning in different sentences: Eat, close, plays	3	5	A	C
7	How does Note-making play an important role in post-reading reflection and brief presentation of thoughts/ ideas/ opinions on the theme of the text.	3	2	C	P
8	Read the follow passage carefully and make notes on it. Also supply a suitable titie to it: A robot can be defined as a mechanical gadget that performs functions normally ascribed to human beings. Karel Capek introduced the word robot while Sir Issac Assimov coined the word robotics which is a science of dealing with robots. The study of robotics includes, selection of material of proper quality for the components, design, fabrication, design of electronic circuits, computers and computer programming and its control. The science of robots is still in developing stage and a lot of research is being pursued for making robots more suitable for working.	3	4	C	F

	Depending upon the area in which robots are to be used, robotics is a multi-dimensional field that includes disciplines such as biology, medical science, psychology, agriculture, mining, various branches of engineering, outer space etc. At present Robots are mainly used in industries. These industrial robots are reprogrammable and perform a variety of jobs through programmed motions. Basically there are two types of robots: fixed and mobile. A fixed robot is attached to an immovable platform. It is similar to a human being standing or sitting in a fixed position while doing the work with hands. On the other hand, a mobile robot moves from place to place. The mobility of a robot is due to wheels or legs or other crawling material provided to it. A mobile robot can be given a human shape. However, the actual shape has nothing to do with real functioning of the robot.				
9	Summarize the opening ceremony of FIFA World Cup, 2022 which took place in Qatar.	3	2		F
10	Mention any four steps involved in planning and preparing a presentation.	3	4		P

Section – C

Attempt All Questions 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	**Read the following text and answer the following questions while making correlations across texts and expressing author's point of view:** The human brain remembers negative experiences more easily than positive ones. Our brains have developed this way because threats, like dangerous animals, had a more immediate effect on our ancestors' survival compared to positive things like food or shelter. As a result, you likely know what makes you unhappy, but do you know what makes you happy? Research suggests that our level of happiness depends partly on factors we cannot control -- our genes and our life circumstances. But our level of happiness is also shaped by the choices we make. If you've been chasing wealth, fame, good looks, material things, and power, you may be looking for happiness in all the wrong places. Psychologists suggest that the following seven habits make people happier. People who form close relationships tend to be happier than those who do not. The number of friends we have is not important. What matters is the quality of our relationships. Relationships that bring happiness usually involve the sharing of feelings, mutual	5	2	U	P

	respect, acceptance, trust, fun and empathy. People who make a habit of caring for the wellbeing of others tend to be happier. This might involve volunteering for an organization or reaching out to support friends and family on a regular basis. People who exercise regularly improve both their physical and mental well-being. Some research has shown that exercise can be as effective as medication in treating depression. When we are so interested in an activity we enjoy that we lose track of time, we are in a state of flow. The activity could be making art, playing piano, surfing, or playing a game. People who experience flow in their work or hobbies tend to be happier. People who include spirituality in their daily life tend to be happier. Practicing spirituality is a way of recognizing and trying to understand the wonder and beauty of existence. Some people do this by going to a place of worship or praying. Some people practice yoga or meditation. Some people go for long walks in nature. People are more likely to be happy if they know what their strengths are and use them regularly. People who set goals and use their strengths to achieve them tend to be happier. People are especially happy when they can use their strengths to serve the greater good. People who think positively by being grateful, mindful, and optimistic are more likely to be happy. Being grateful means being thankful. Being mindful means being open to, focusing on, and enjoying the experiences of the present moment. Being optimistic means being hopeful about the future. Ques 1: How does the author explain spirituality? Ques 2: Why doesn't the brain remember positive experiences as much as negative ones? Ques 3: What is not part of being mindful? Ques 4: What would the author think is not a way of caring for the well-being of others? Ques 5: How can a person overcome depression?				
12	As the participant of a group discussion, You are required to speak on the topic, "Top Tech Layoffs At Big Tech Companies in 2022". Justify your stand with suitable examples. Or Discuss the importance of Interview Skills. Support your answer with example.	5	5	E	C

Course Name: Disaster Management

Course Outcome

CO1- Understanding foundations of hazards, disasters and associated natural/social phenomena

CO2- Familiarity with disaster management theory (cycle, phases)

CO3- Knowledge about existing global frameworks and existing agreements

CO4- Methods of community involvement as an essential part of successful DRR

CO5- Technological innovations in Disaster Risk Reduction: Advantages and problems

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End Term Examination, Odd Semester, Session 2022-23
B. Tech (CE, CS, ME, MTE), 3rd Semester, 2nd Year
BCHM0101 & Disaster Management

Time: 1 Hours **Maximum Marks: 100**

Attempt All Questions 2 X 50 = 100 Marks

1. Hazards and Disasters are mainly classified as?
   a) Physical and Chemical
   b) Natural and Human Induced
   c) Physical and Human
   d) Social and Cultural
2. Which of the following is most likely to be in the 'eye' of a cyclone?
   a) It is an area of high pressure
   b) It is an area of low pressure
   c) It has high speed winds
   d) It has lots of clouds
3. One of the main reasons for farmers commit suicide in India is
   a) Crop failure
   b) Earthquake
   c) Tsunami
   d) All of the above
4. Disaster is an event arising out of
   a) Result of hazard event
   b) Causes of hazard event
   c) Causes of disaster event
   d) All of the above
5. The Richter scale expresses an earthquakes
   a) Magnitude
   b) Location
   c) Duration
   d) Depth
6. Earthquakes occur when there is a sudden release of stored up energy in Earth's
   a) Inner core
   b) Outer core
   c) Upper mantle
   d) Lower crust
7. During a tsunami a series of waves radiating out ward from a central disturbance of the sea floor is called
   a) Ocean storm
   b) Wind-generated disturbance
   c) Wave train
   d) Wavelets
8. Volcanic erupted material when inside the hill/earth/mountain it is called
   a) Lava
   b) Magma
   c) Lahars
   d) None of these
9. A hazard is a situation where there is?
   a) Threat of natural calamity
   b) Threat to property and lives from calamities
   c) Threat for consequences of disaster
   d) All of the above
10. The point of the earth's surface directly above the point where an earthquake occurs is called
    a) Focus
    b) Epicenter
    c) Fracture
    d) Fault
11. Generally the number on Richter Scale ranges between –
    a) 0 and 6
    b) 0 and 9
    c) 1 to 5
    d) 1 to 12

12. From where earth quake waves are generated?
    a) Focus
    b) Epicenter
    c) Solid inner Core
    d) None of these
13. What are the three phases of disaster management planning?
    a) Preparation, Response and Recovery
    b) Preparation, Planning and Perception
    c) Evacuating, Rebuilding and Re-branding
    d) Planning, Evacuating and Recovery
14. Which city burned in the "Great Fire" of 1666?
    a) London
    b) Rome
    c) Paris
    d) Berlin
15. Why does site history have to be considered for hazard identification?
    a) To estimate the risk
    b) To calculate carcinogenic exposure
    c) To know the probable source and causes of contamination on site
    d) For determination of remedial actions
16. Responsibility for securing the scene, preserving life and treating the wounded is th responsibility of
    a) First responders
    b) Mental health professionals
    c) Social workers who specialize in crisis management
    d) All citizens
17. Where did the strongest earthquake ever recorded occur?
    a) Japan
    b) USA
    c) Chile
    d) India
18. In 1952, what disaster in London lasted for five days and killed roughly 12,000 people?
    a) Acid rain storm
    b) Polluted fog
    c) River Thames over flowing
    d) Blizzard
19. In northern hemisphere, the wind of the tropical cyclone blows in
    a) Anticlockwise direction
    b) Clockwise direction
    c) Straight
    d) None of the above
20. A flash flood is a flood that
    a) Is caused by heavy rain rather than from the flooding of a river
    b) Occurs in urban areas
    c) Occurs suddenly and unexpectedly and for a short duration
    d) Is caused by the blocking of drains.
21. In which Phase of Disaster damage Assessment is done?
    a) Warning phase
    b) Impact Phase
    c) Rescue phase
    d) Re-Habilitation Phase
22. Below average rainfall for a prolonged period of time is called
    a) Famine
    b) Drought
    c) Flood
    d) None of these
23. About 2/3 rd of the cyclones that occur in the Indian coastline occur in the
    a) Bay of Bengal
    b) Coastal area of south India
    c) Coastal area of west India
    d) None of the above
24. Vulnerability analysis comes in which part of the Disaster Management Cycle
    a) Mitigation
    b) Preparedness
    c) Response
    d) Recovery

25. Which event produces the biggest tsunami?
    a) Earthquake
    b) Underwater landslides
    c) Hurricanes
    d) Impacts of asteroids and comets
26. What cloud type can form a tornado?
    a) Cirrus
    b) Cumulus
    c) Cumulonimbus
    d) Stratus
27. Which of the following data is not required for hazard identification?
    a) Land use
    b) Contaminant levels
    c) Affected population
    d) Estimation of risk
28. Covid19 is due to a virus named
    a) MERS-CoV
    b) SARS-CoV
    c) MERS-CoV2
    d) SARS-CoV2
29. The Plan prepared for facilitating the prompt return to the normalcy and continuity of operation after a disaster is called:
    a) Rescue Plan
    b) Relief Plan
    c) Recovery Plan
    d) Response Plan
30. The center of a cyclone is a calm area and is called the storm.
    a) Eye
    b) Focus
    c) Centre
    d) Radius
31. A disease that becomes unusually widespread and even global in its reach is referred to as
    a) Epidemic
    b) Pandemic
    c) Spanish flu
    d) Hyper endemic
32. Which of the following is not a man-made hazard?
    a) Leakage of Toxic waste
    b) Wars and Civil Strife
    c) Drought
    d) Environmental pollution
33. The destructive powers of tsunami result mainly from their _____.
    a) incredible height
    b) unpredictability
    c) cold water
    d) momentum and long wavelength
34. Among which of the following Hazards and Disasters are classified?
    a) Cultural and social
    b) Natural and man-made
    c) Chemical and physical
    d) Human and physical
35. What does "tsunami" mean in Japanese?
    a) Severe earthquake
    b) Huge wind
    c) Harbour wave
    d) Terrible storm
36. A cyclone is called a_____ in Philippines and Japan.
    a) Hurricane
    b) Typhoon
    c) Tornado
    d) Thunderstorm
37. Tsunamis are most commonly found in which ocean?
    a) Pacific Ocean
    b) Atlantic Ocean
    c) Artic Ocean
    d) Indian Ocean

38. The National Disaster Management Authority (NDMA) is headed by
    a) Prime Minister of India
    b) President of India
    c) Governor of States
    d) Chief Minister of States
39. Which one of the following does not cause Earthquake?
    a) Colliding plates
    b) Coral reefs
    c) Sliding plates
    d) Dividing plates
40. Which of the following agencies are the National Disaster Risk Reduction stakeholders?
    a) Central Government
    b) National Disaster Offices
    c) Private Sector
    d) All of the above
41. What are the major types oriented to mitigation measures of hazards and disasters?
    a) Public mitigation measures
    b) Community mitigation measures
    c) Both a and b
    d) None of the above
42. The steps need to be taken for any risk natural or man-made disaster
    a) Proper building construction
    b) A well planned
    c) Awareness among the stakeholders
    d) All of the above
43. Disaster management is aimed at
    a) Resettling people in the closest unaffected urban area
    b) Collection of valuable data for future management objectives
    c) Strengthening sewage and drinking water treatment facilities to resist the impact of a future disaster
    d) Restoring a communities' services, facilities and residences to pre-disaster level
44. What are the major Hazard/disaster management related activities?
    a) Preparedness
    b) Response
    c) Recovery
    d) All of the above
45. The NH-44 (presently 8) is affected by which hazard
    a) Floods
    b) Earthquake
    c) Landslide
    d) All of the above
46. Who shall be the first responders in disaster management situation?
    a) Central Government
    b) Community
    c) Local Government
    d) State Government
47. Example of anthropogenic hazards are:
    a) Chemical explosion
    b) Earthquake
    c) Flood
    d) Wild fire
48. The Bhopal gas tragedy is an example of:
    a) Industrial disasters
    b) Natural disasters
    c) Nuclear disasters
    d) None
49. The primary goal of a disaster preparedness plan is:
    a) To protect the population
    b) To protect valuable resources
    c) To keep communications lines open
    d) To protect environmental health personnel
50. Landslides occur because of?
    a) Intensity of rainfall
    b) Steep slopes
    c) Deforestation leading it soil erosion
    d) All the above

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End Term Examination, Odd Semester 2022-23
B. Tech. (CS-All Branch), II Year, III Semester
BALM 0001 : Indian Constitution

Time: 1 Hour **Maximum Marks: 100**

Attempt All Questions 2 X 50 = 100 Marks

1. The Governor of a State is appointed by the President on the advice of the
(a) Prime Minister (c) Chief Minister
(b) Vice- President (d) Chief Justice
2. The President gives his resignation to the
(a) Chief Justice (c) Vice President
(b) Parliament (d) Prime Minister
3. For what period does the Vice President of India hold office ?
(a) 5 years (c) 6 years
(b) Till the age of 65 years (d) 2 years
4 Who among the following holds office during the pleasure of the President ?
(a) Governor (c) Speaker of Lok Sabha
(b) Election Commissioner (d) Prime Minister
5. Which of the following is not true regarding the payment of the emoluments of the President ?
(a) They can be reduced during a Financial Emergency. (c) They are charged on the Contigency Fund of India.
(b) They are shown separately in the budget. (d) They do not require any parliament sanction.
6. The total number of members nominated by the President to the Lok Sabha and the Rajya Sabha is
(a) 16 (c) 14
(b) 18 (d) 12
7. Which one of the following does not constitute the electoral college for electing the President of India ?
(a) Elected members of Lok Sabha (c) Elected members of the Legislative Council
(b) Elected members of the Legislative Assembly of each state. (d) Elected members of Rajya Sabha
8. The council of Ministers in a Parliamentary type of Government can remain in office till it enjoys the support of the
(a) Minority of members of the Upper House of Parliament
(b) Majority of the members of the Upper House of Parliament
(c) Minority of members of the Lower House
(d) Majority of the members of the Lower House of Parliament
9. Which of the following appointments is not made by the President of India ?
(a) Chief of the Army (c) Chief Justice of India
(b) Speaker of the Lok Sabha (d) Chief of the Air Force
10. Who appoints the Prime Minister of India ?
(a) Lok Sabha (c) Parliament
(b) President (d) Citizens of India
11. The first woman Governor of a state in free India was
(a) Mrs. Indira Gandhi (c) Mrs. Sarojini Naidu
(b) Mrs. Vijaya Laxmi Pandit (d) Mrs. Sucheta Kripalani
12. Minimum age required to contest for Presidentship is
(a) 23 years (c) 35 years
(b) 21 years (d) 30 years

13. The charge of impeachment against the President of India for his removal can be preferred by
(a) Both Houses of Parliament
(b) Speaker of Lok Sabha and Chairman of Rajya Sabha
(c) Rajya Sabha
(d) Lok Sabha

14. The Chief Minister of a Union Territory where such a set up exists, is appointed by the
(a) Lt. Governor
(b) Majority party in Legislature
(c) President
(d) Prime Minister

15. Who was the first Prime Minister of India ?
(a) Jawaharlal Nehru
(b) Mrs. Indira Gandhi
(c) Dr. Rajendra Prasad
(d) Mahatma Gandhi

16. In case a President dies while in office, the vice President can act as President for a maximum period of
(a) 1 years
(b) 3 months
(c) 6 months
(d) 2 years

17. The Union Council of Ministers consists of
(a) Cabinet Ministers, Minister of State and Deputy Ministers
(b) Cabinet Ministers and Chief Ministers of the States
(c) Prime Minister
(d) Cabinet Ministers

18. Who administers the oath of office to the President of India before he enters upon the office ?
(a) Chief Justice
(b) Speaker
(c) Vice President
(d) Prime Minister

19. Who among the following enjoys the rank of a Cabinet Minister of the Indian Union?
(a) None of the Above
(b) Deputy Chairman of the Planning Commission
(c) Deputy Chairman, Rajya Sabha
(d) Secretary to the Government of India

20. A person who is not a member of Parliament can be appointed as a Minister by the President for a maximum period of
(a) 9 months
(b) 3 months
(c) 12 months
(d) 6 months

21. When can a President use his discretion in appointing the Prime Minister ?
(a) When no political oarty enjoys a clear majority in the Lok Sabha.
(b) Only when the Lok Sabha has been dissolved.
(c) In all circumstances
(d) In no circumstances

22. If in a country there happens to be the office of monarchy as well as a Parliamentary form of government this monarch will be called
(a) Head of the State
(b) Head of the Cabinet
(c) Head of the government
(d) Both Head of the government and State

23. One feature distinguishing the Rajya Sabha from the Vidhan Parishad is
(a) Power of impeachment
(b) Indirect election
(c) Nomination of members
(d) Tenure of membership

24. Chief Ministers of all the states are ex-officio members of the
(a) Planning Commission
(b) National Development Council
(c) Inter State Council
(d) Finance Commission

25. Who, among the following, has the final right to sanction the expenditure of public money in India ?
(a) Speaker
(b) President
(c) Prime Minister
(d) Parliament

26. Which of the following is not an essential qualification for appointment as a Governor ?
(a) He must not be a member of either House of Parliament.
(b) He should be a domicile of the state to which he is being appointed.
(c) He should be a citizen of India.
(d) He must have completed the age of 35 years

27. The President nominates 12 members of the Rajya Sabha according to
(a) Their performance as office bearers of cultural societies.
(b) The recommendations made by the Vice President.
(c) Their role played in political set up of the country.
(d) Their distinction in the field of science, art, literature and social service.
28. Which Article of the Constitution empowers the President to appoint a Commission to investigate the condition of backward classes in general and suggest ameliorative measures?
(a) Art 342
(b) Art 344
(c) Art 340
(d) Art 339
29. Who can initiate impeachment proceedings against the President of India ?
(a) Either House of Parliament
(b) Any Vidhan Sabha
(c) Only Lok Sabha
(d) Rajya Sabha
30. Who was the member of the Rajya Sabha when first appointed as the Prime Minister of India?
(a) Lal Bahadur Shastri
(b) Charan Singh
(c) Morarji Desai
(d) Indira Gandhi
31. What is the minimum age for appointment as a Governor ?
(a) 35 years
(b) 40 years
(c) 25 years
(d) 30 years
32. Chief Minister of a State is responsible to
(a) Prime Minister
(b) Rajya Sabha
(c) Legislative Assembly
(d) Governor
33. The administrative and operational control of the Armed Forces is exercised by the
(a) Cabinet Committee on Political Affairs with Prime Minister as the Chairman
(b) Ministry of Defence
(c) Three Chiefs of the Army, Navy and Air Force Staff
(d) President
34. What is the position of a Minister of State in the Central Government ?
(a) He is a Minister of Central Government but not a member of the Cabinet.
(b) He looks after the interests of the State Cabinet.
(c) He is the nominee of the State Governor.
(d) He is the nominee of the State Cabinet.
35. Who among the following is directly responsible to Parliament for all matters concerning the Defence Services of India ?
(a) President
(b) Prime Minister
(c) Defence Minister
(d) Cabinet Committee on Political Affairs
36. How many times the President of India can seek re-election to his post ?
(a) Once
(b) 3 times
(c) 2 times
(d) Any number of times
37. Who among the following has the power to form a new state within the Union of India ?
(a) Speaker of Lok Sabha
(b) President
(c) Prime Minister
(d) Supreme Court
38. Is the Prime Minister bound to advise the President on matters on which his advice is sought?
(a) If the Council of Ministers so desires.
(b) Yes
(c) No
(d) It is discretionary
39. When the Vice President officiates as President he draws the salary of
(a) Chairman of Rajya Sabha
(b) Both (A) and (C)
(c) President
(d) Member of Parliament
40. Who among the following can attend meetings of the Union Cabinet ?
(a) President
(b) Cabinet Ministers
(c) Ministers of State
(d) Deputy Ministers

41. Who among the following has the Constitutional authority to make rules and regulations fixing the number of members of the UPSC
(a) Vice- President
(b) President
(c) Home Ministry
(d) Cabinet Secretary

42. Under whose advice the President of India declares Emergency under Article 352 ?
(a) Chief Ministers of all states
(b) Prime Minister
(c) Cabinet
(d) Council of Ministers

43. Who acts as the President of India when neither the President nor the Vice President is available ?
(a) Seniormost Governor of a State
(b) Chief Justice of India
(c) Speaker of Lok Sabha
(d) Auditor General of India

44. A proclamation of emergency issued under Article 36 must be approved by the Parliament within
(a) 3 months
(b) 2 months
(c) 1 Month
(d) 6 weeks

45. The Union Council of Ministers is appointed by the
(a) President of India according to his discretion
(b) President of India on the advice of the Prime Minister
(c) Prime Minister of India
(d) Parliament

46. The President can proclaim emergency on the written advice of the
(a) Speaker of Lok Sabha
(b) Prime Minister
(c) Chief Justice of India
(d) Union Cabinet

47. The Union Council of Ministers is collectively responsible to
(a) Rajya Sabha
(b) President
(c) House of the People
(d) Prime Minister

48. Article 78 of the Constitution deals with
(a) President's power to get information from the Council of Ministers.
(b) Prime Minister's duty regarding keeping the President informed about the government's decisions and policies.
(c) Emergency powers of the President
(d) President's power to send advisory messages to the Parliament.

49. The maximum duration for which the Vice President may officiate as the President, is
(a) 1 year
(b) 6 months
(c) 4 months
(d) 9 months

50 The President's Rule in a State can be continued for a maximum period of
(a) 2 years
(b) 1 year
(c) 6 months
(d) 2 1/2 years